

- 7 The photograph shows a mother and baby Malayan tapir (*Tapirus indicus*). They live in the rainforests of southeast Asia.

Each gamete of a Malayan tapir contains 26 chromosomes.



© Sipa US / Alamy Stock Photo

- (a) The body cells of the mother Malayan tapir have some genetic similarities to and differences from the cells of her baby.

Explain why the body cells of the mother and baby have genetic similarities **and** differences.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....



- (b) Biodiversity within a species can be measured using genetic diversity.

Three melanistic Malayan tapirs have been discovered that have black fur all over their bodies.

Melanistic tapirs have two copies of the recessive allele for fur colour and are homozygous.

- (i) Scientists wanted to determine the heterozygosity index of the Malayan tapir species.

Which equation can be used to calculate the heterozygosity index?

(1)

Equation	
☐ A	heterozygosity index = $\frac{\text{number of heterozygotes}}{\text{number of individuals in the population}}$
☐ B	heterozygosity index = $\frac{\text{number of heterozygotes}}{\text{number of species in the habitat}}$
☐ C	heterozygosity index = $\frac{\text{number of individuals in the population}}{\text{number of heterozygotes}}$
☐ D	heterozygosity index = $\frac{\text{number of species in the habitat}}{\text{number of heterozygotes}}$

- (ii) Explain how scientists could determine if a **change** in the frequency of the recessive allele was occurring over time.

(2)

.....

.....

.....

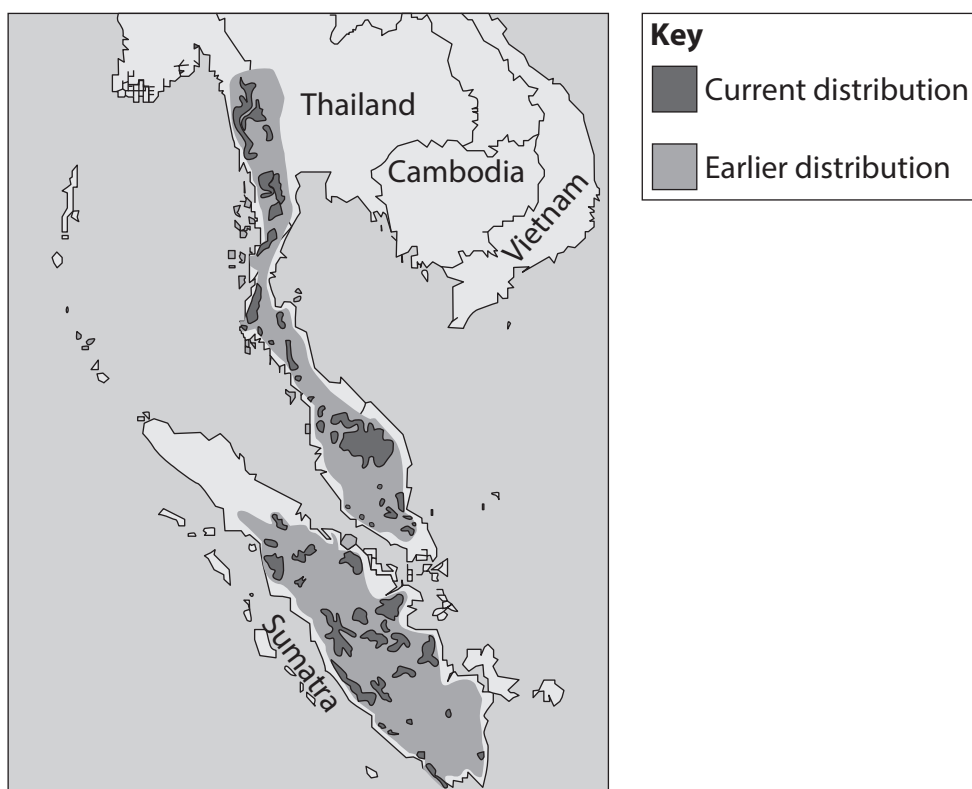
.....

.....

.....



- (c) The map shows the change in distribution of the Malayan tapir over the past 20 years.



- (i) Give **one similarity** between the current and earlier distribution of the Malayan tapir.

(1)

- (ii) One difference is that the distribution has reduced due to human activity.

Suggest **one** reason why human activity has caused this change in distribution.

(1)



- (iii) A study suggested that the genetic diversity of the Malayan tapirs has decreased due to populations being separated from each other.

Explain how a captive breeding programme could maintain the genetic diversity of Malayan tapir populations.

(3)

(Total for Question 7 = 11 marks)

